

Coordinate-free Mathematics

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1 Algebra

1.1 Linear algebra

Fix a field k . A k -**vector space** is an abelian group $(V, +)$ with a scalar multiplication $k \times V \rightarrow V$ satisfying

$$\begin{aligned}\alpha(v + w) &= \alpha v + \alpha w, & (\alpha + \beta)v &= \alpha v + \beta v, \\ (\alpha\beta)v &= \alpha(\beta v), & 1v &= v\end{aligned}$$

for $\alpha, \beta \in k$ and $v, w \in V$. A **linear map** $f : V \rightarrow U$ between k -vector spaces satisfies

$$f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$$

for $\alpha, \beta \in k$ and $v, w \in V$.

Exercise 1 (Small string vibrations obey superposition)

Derive the wave equation starting from a taut string and show that its solutions form a vector space. Show that time evolution from initial data to the state at time t is a linear map.

The **free k -vector space on a set X** , denoted kX , is the set of finitely supported set-theoretical functions $X \rightarrow k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W , denoted V/W , is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is $k(V \times W)/N$, where N is spanned by

$$\begin{aligned}(v + v', w) - (v, w) - (v', w), & \quad (v, w + w') - (v, w) - (v, w'), \\ (\lambda v, w) - \lambda(v, w), & \quad (v, \lambda w) - \lambda(v, w)\end{aligned}$$

for $v, v' \in V$, $w, w' \in W$, and $\lambda \in k$.

An **isomorphism** of vector spaces $V \cong W$ consists of linear maps $f : V \rightarrow W$ and $g : W \rightarrow V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$.

Exercise 2 (Universal constructions)

Give the universal properties of free spaces, direct sums, quotients, and tensor products, and use them to prove the three isomorphism theorems and $k(X \sqcup Y) \cong kX \oplus kY$, $k(X \times Y) \cong kX \otimes kY$.

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k . The evaluation map $\text{ev}_V : V^* \otimes V \rightarrow k$ is the linear extension of the evaluation morphism $f \otimes x \mapsto f(x)$. We say that V is **finite dimensional** if there is a co-evaluation morphism $\text{coev}_V : k \rightarrow V \otimes V^*$ satisfying the snake equations $(1_V \otimes \text{ev}_V) \circ (\text{coev}_V \otimes 1_V) = 1_V$ and $(\text{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \text{coev}_V) = 1_{V^*}$. We define the **dimension** $\dim(V) = \text{tr}(1_V)$, where $\text{tr} : \text{End}(V) \rightarrow k$ is given by $f \mapsto (\text{ev}_V \circ s_{V, V^*} \circ (f \otimes 1_{V^*}) \circ \text{coev}_V)(1)$ where $s_{X, Y} : X \otimes Y \rightarrow Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$.

Exercise 3 (Trace and dimension identities)

Prove the following for finite-dimensional vector spaces V, W .

- If $f : V \rightarrow W$ and $g : W \rightarrow V$, then $\text{tr}(gf) = \text{tr}(fg)$.
- If $f \in \text{End}(V)$ and $g \in \text{End}(W)$, then $\text{tr}(f \oplus g) = \text{tr}(f) + \text{tr}(g)$.
- If $f \in \text{End}(V)$ and $g \in \text{End}(W)$, then $\text{tr}(f \otimes g) = \text{tr}(f) \text{tr}(g)$.
- If $\text{char}(k) = 0$, then $\dim(V) \in \mathbf{N} \subset k$
- If $\text{char}(k) = 0$, then $V \cong W$ iff $\dim(V) = \dim(W)$

Define the tensor algebra and **exterior algebra of V** by

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \quad \Lambda V = T(V) / \langle v \otimes v : v \in V \rangle.$$

The space $\Lambda^k V$ is the degree- k component of ΛV .

Exercise 4 (The top exterior power is one-dimensional) Let V be finite-dimensional with $n = \dim(V)$. Show that $\Lambda^n V$ is one-dimensional and that $\Lambda^k V = 0$ for all $k > n$.

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by $v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots \wedge f(v_{\dim(V)})$.

Exercise 5 (Determinant identities and volume)

Prove the following:

- $\det(f \oplus g) = \det(f) \det(g)$
- If $V = \mathbb{R}^n$, then $\det(f)$ is the signed volume of $f(I^n)$ where $I = [0, 1]$.
- Cramer's rule

For $f \in \text{End}(V)$, the **characteristic polynomial** is the polynomial $\lambda \mapsto \det(f - \lambda 1_V)$.

A scalar $\lambda \in k$ is an **eigenvalue** of f if $\ker(f - \lambda 1_V) \neq 0$. This kernel is its **eigenspace**; its dimension is the geometric multiplicity of λ .

Exercise 6 (Characteristic polynomial and Jordan form)

Let $f \in \text{End}(V)$ have split characteristic polynomial. Prove the Cayley-Hamilton theorem and Jordan normal form, and derive the algebraic and geometric multiplicity inequality together with the eigenvalue formulas for $\text{tr}(f)$ and $\det(f)$.

Exercise 7 (Critical damping is a Jordan block)

Use Jordan normal form to solve $m\ddot{x} + b\dot{x} + \kappa x = 0$ in all three damping regimes, and show that critical damping is exactly $b^2 = 4m\kappa$, where a nontrivial Jordan block produces $te^{-bt/(2m)}$.

A **Hermitian inner product** on a complex vector space V is a positive-definite sesquilinear form $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable and linear in the second. It induces the norm $\|v\| = \sqrt{\langle v, v \rangle}$. For a linear map $A : V \rightarrow W$ between finite-dimensional Hermitian spaces, an **adjoint** is a map $A^* : W \rightarrow V$ satisfying

$$\langle Av, w \rangle = \langle v, A^*w \rangle.$$

An endomorphism A is **self-adjoint** if $A = A^*$, **normal** if $A^*A = AA^*$, **positive** if $\langle v, Av \rangle \geq 0$ for every v , and **unitary** if $A^*A = AA^* = 1$. Equip $\text{End}(V)$ with the induced operator norm.

Exercise 8 (Finite-dimensional spectral theorem)

Develop the finite-dimensional spectral theorem: prove existence of adjoints and positive square roots, and prove that every normal operator has an orthogonal spectral decomposition

$$A = \sum_{\lambda} \lambda p_{\lambda}, \quad \sum_{\lambda} p_{\lambda} = 1.$$

Deduce the singular-value decomposition

$$T = U\Sigma V^*.$$

Exercise 9 (The spectral decomposition produces normal modes)

For a fixed-end chain of N equal masses and springs, set $q_0 = q_{N+1} = 0$ and define

$$(Kq)_j = \frac{\kappa}{m}(2q_j - q_{j-1} - q_{j+1}).$$

Find the spectral decomposition of K and use it to determine all normal modes of $\ddot{q} + Kq = 0$, their frequencies, and the conserved energy

$$\mathcal{E}(q, \dot{q}) = \frac{m}{2} \|\dot{q}\|^2 + \frac{m}{2} \langle q, Kq \rangle.$$

Exercise 10 (Fourier modes solve the fixed-end wave equation)

Solve the fixed-end wave equation from Exercise 1 by Fourier series, prove conservation of wave energy, and recover its normal-mode frequencies as the continuum limit of Exercise 9.

1.2 Group representations

An action of a group G on a set X assigns to each $g \in G$ a bijection $x \mapsto gx$ such that $1x = x$ and $(gh)x = g(hx)$. A **representation** of G on a vector space V is a homomorphism

$$\rho : G \longrightarrow \text{GL}(V).$$

A subspace $W \subseteq V$ is **G -invariant** if $\rho(g)W \subseteq W$ for every $g \in G$. Restriction gives the **subrepresentation** W , and $\rho(g)(v + W) = \rho(g)v + W$ gives the **quotient representation** V/W . A nonzero representation is **irreducible** if its only invariant subspaces are 0 and V .

For representations V and W , the direct sum, tensor product, and dual representations are defined by

$$\begin{aligned} g(v, w) &= (gv, gw), \\ g(v \otimes w) &= gv \otimes gw, \\ (g\phi)(v) &= \phi(g^{-1}v), \quad \phi \in V^*. \end{aligned}$$

An **intertwiner** $T : V \rightarrow W$ is a linear map satisfying $T\rho_V(g) = \rho_W(g)T$ for every g . Write $\text{Hom}_G(V, W)$ for the space of intertwiners and

$$V^G = \{v \in V : \rho(g)v = v \text{ for every } g \in G\}$$

for the fixed-point space.

Exercise 11 (Averaging produces invariants)

Let G be finite and let V be a complex representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

Prove that P_G is the projection of V onto V^G . Prove also that averaging any Hermitian inner product over G produces a G -invariant Hermitian inner product, so every finite-dimensional complex representation of G is equivalent to a unitary representation.

1.3 Maschke's theorem and Schur's lemma

Exercise 12 (Maschke's theorem)

Let $W \subseteq V$ be G -invariant and let $P : V \rightarrow W$ be any linear projection. Define

$$\tilde{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g)P\rho(g)^{-1}.$$

Prove that $\tilde{P}$ is G -equivariant and has image W . Deduce that $\ker \tilde{P}$ is a G -invariant complement of W . If G is finite and the characteristic of k does not divide $|G|$, prove that every finite-dimensional k -representation of G is a direct sum of irreducible representations.

Exercise 13 (Schur's lemma)

Let V and W be irreducible finite-dimensional complex representations. Prove that every nonzero intertwiner $T : V \rightarrow W$ is an isomorphism and that

$$\text{End}_G(V) = \mathbb{C}1_V.$$

Let $\mathbb{C}[G]$ act on V by the linear extension of ρ . Deduce that every element of the center $Z(\mathbb{C}[G])$ acts by a scalar on each irreducible representation.

1.4 Characters and orthogonality

The **character** of a finite-dimensional representation V is the function

$$\chi_V(g) = \text{tr}(\rho_V(g)).$$

For class functions $\chi, \psi : G \rightarrow \mathbb{C}$, define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

Exercise 14 (Great Orthogonality Theorem)

For irreducible unitary representations Γ^α and Γ^β of dimensions d_α, d_β of a finite group G , prove

$$\sum_{g \in G} \Gamma^\alpha(g)_{ij} \overline{\Gamma^\beta(g)_{kl}} = \frac{|G|}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{jl}.$$

Develop its character consequences, including

$$m_\alpha = \langle \chi_\alpha, \chi_V \rangle_G.$$

Deduce that characters classify finite-dimensional complex representations.

For an irreducible character χ_α , define the **symmetry projection**

$$P_\alpha = \frac{d_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho(g).$$

Exercise 15 (Symmetry projections)

Prove that the operators P_α are pairwise orthogonal idempotents and that $P_\alpha V$ is the α -isotypic component of V . Interpret P_α as constructing symmetry-adapted linear combinations of atomic orbitals or displacement vectors.

1.5 Molecular symmetry

The **point group** of a molecule is the finite group $G \subseteq O(3)$ of spatial symmetries preserving its nuclear configuration, including atomic species. For a molecule with N atoms, G acts on the real displacement space

$$V_{\text{disp}} = \bigoplus_{a=1}^N \mathbb{R}_a^3 \cong \mathbb{R}^{3N}$$

by permuting atoms and applying the spatial linear action to each displacement. The three-dimensional translational representation carries the polar-vector action R_g , while the rotational representation carries the axial-vector action $\det(R_g)R_g$; denote them by V_{trans} and V_{rot} . We complexify these real representations

when applying character theory. For a nonlinear molecule, the vibrational representation is the class

$$[V_{\text{vib}}] = [V_{\text{disp}}] - [V_{\text{trans}}] - [V_{\text{rot}}]$$

in the representation ring.

Exercise 16 (Character of molecular displacements)

For $g \in G$, prove that

$$\chi_{\text{disp}}(g) = \sum_{g(a)=a}^a \text{tr}(R_g),$$

where $R_g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the spatial action and the sum is over atoms fixed by g . Prove that $\dim V_{\text{vib}} = 3N - 6$ for a nonlinear molecule, and use character orthogonality to express the multiplicity of every irreducible representation in V_{vib} .

For water, take the z -axis along the C_2 axis and the molecule in the yz -plane. Its point group is

$$C_{2v} = \{E, C_2, \sigma_{xz}, \sigma_{yz}\}.$$

Exercise 17 (Water has three symmetry-classified vibrations)

Derive the C_{2v} character table and use it to show that water's vibrational representation is

$$V_{\text{vib}} \cong 2A_1 \oplus B_2.$$

Identify and construct the three normal modes.

1.6 Spectroscopic selection rules

Let V_i and V_f be the symmetry types of initial and final states. The electric-dipole operator transforms as the vector representation $V_\mu = V_{\text{trans}}$. A transition amplitude

$$\langle \psi_f, \mu \psi_i \rangle$$

is the matrix coefficient used below.

Exercise 18 (Symmetry determines electric-dipole transitions)

Derive the electric-dipole selection rule

$$1 \subseteq V_f^* \otimes V_\mu \otimes V_i.$$

Apply it to show that all three fundamental vibrations of water are infrared active. Compare the mode assignments with the compiled gas-phase fundamentals

$$\nu_2 = 1594.59 \text{ cm}^{-1}, \quad \nu_1 = 3656.65 \text{ cm}^{-1}, \quad \nu_3 = 3755.79 \text{ cm}^{-1}$$

for the bend, symmetric stretch, and asymmetric stretch [35]. Reconcile the compilation's B_1 label for the asymmetric stretch with the B_2 label above by comparing axis conventions.

Exercise 19 (Water's vibrations are Raman active)

Let the polarizability tensor carry $\text{Sym}^2(V_\mu)$. Prove that a vibrational mode is Raman active precisely when its irreducible representation occurs in this symmetric square. Prove that

$$\text{Sym}^2(V_\mu) \cong 3A_1 \oplus A_2 \oplus B_1 \oplus B_2$$

for C_{2v} . Deduce that all three fundamental vibrational modes of water are Raman active.

1.7 Lie groups, infinitesimal symmetry, and spin

The molecular point groups above are discrete subgroups of $O(3)$. Define the group of orientation-preserving spatial rotations by

$$\text{SO}(3) = \{R \in M_3(\mathbb{R}) : R^T R = 1, \det R = 1\}.$$

A **matrix Lie group** is a subgroup $G \subseteq \text{GL}(n, \mathbb{C})$ that is closed in the subspace topology. Its **Lie algebra** is

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) : e^{tX} \in G \text{ for all } t \in \mathbb{R}\},$$

where $e^{tX} = \sum_{k \geq 0} (tX)^k / k!$. The Lie bracket is $[X, Y] = XY - YX$. An abstract **Lie algebra** over k is a k -vector space with a bilinear bracket that is skew-symmetric and satisfies the Jacobi identity. A **Lie algebra representation** on V is a linear map $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ with $\rho([X, Y]) = [\rho(X), \rho(Y)]$.

For $g \in G$ and $X, Y \in \mathfrak{g}$, the adjoint actions are

$$\text{Ad}_g(X) = gXg^{-1}, \quad \text{ad}_X(Y) = [X, Y].$$

Exercise 20 (Classical matrix Lie algebras)

Prove that the Lie algebras of $\text{GL}(n, \mathbb{C})$, $\text{SL}(n, \mathbb{C})$, $\text{O}(n)$, $\text{SO}(n)$, $\text{U}(n)$, and $\text{SU}(n)$ are respectively

$$\mathfrak{gl}(n, \mathbb{C}), \quad \mathfrak{sl}(n, \mathbb{C}) = \{X : \text{tr}(X) = 0\},$$

$$\mathfrak{o}(n) = \{X : X^T = -X\}, \quad \mathfrak{u}(n) = \{X : X^* = -X\}, \quad \mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n, \mathbb{C}).$$

Verify that each is a Lie algebra under $[X, Y] = XY - YX$.

Exercise 21 (One-parameter subgroups and the bracket)

Prove that $t \mapsto e^{tX}$ is a smooth group homomorphism $\mathbb{R} \rightarrow G$ whenever $X \in \mathfrak{g}$, and that every smooth homomorphism $\mathbb{R} \rightarrow G$ arises uniquely in this way. Prove that

$$[X, Y] = \left. \frac{d}{dt} \right|_{t=0} \left. \frac{d}{ds} \right|_{s=0} e^{tX} e^{sY} e^{-tX} e^{-sY},$$

and deduce $\text{Ad}_{e^{tX}} = e^{t \text{ad}_X}$.

Exercise 22 (Differentiating group representations)

Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a smooth finite-dimensional representation of a matrix Lie group. Prove that

$$\dot{\rho}(X) = \left. \frac{d}{dt} \right|_{t=0} \rho(e^{tX})$$

is a Lie algebra representation $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$, and that $\rho(e^{tX}) = e^{t\dot{\rho}(X)}$. If G is connected, prove that a subspace is G -invariant if and only if it is $\mathfrak{g}$ -invariant.

Exercise 23 (Rotations, angular momentum, and spin)

Using the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Identify $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, construct the double cover $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$, and classify the irreducible spin representations $\rho_j : \mathrm{SU}(2) \rightarrow U(V_j)$. For $J_a = i\dot{\rho}_j(-i\sigma_a/2)$, derive $\dim V_j = 2j + 1$ and $J^2 = j(j + 1)1_{V_j}$, including the distinction between integer and half-integer spin.

A **quantum number** is an eigenvalue used to label a joint eigenspace of commuting observables.

Exercise 24 (Energy and symmetry produce quantum numbers)

Let H be a self-adjoint operator on a finite-dimensional Hermitian space V with a commuting $\mathrm{SU}(2)$ action. Derive the joint quantum-number decomposition

$$|E, j, m, \alpha\rangle$$

for H, J^2, J_3 , including the allowed values and the degeneracy forced by spin j .

Exercise 25 (Hidden symmetry gives the hydrogen spectrum)

With $p = -i\hbar\nabla$, $r = |x|$, and $L = x \times p$, consider the hydrogen Hamiltonian and Runge–Lenz operator

$$H = \frac{p^2}{2\mu} - \frac{\kappa}{r}, \quad A = \frac{1}{2\mu}(p \times L - L \times p) - \kappa \frac{x}{r}.$$

Use the hidden $\mathfrak{so}(4)$ symmetry to derive the bound-state spectrum and degeneracies

$$E_n = -\frac{\mu\kappa^2}{2\hbar^2 n^2}, \quad n \geq 1, \quad \dim W_n = n^2,$$

and the angular-momentum decomposition

$$W_n \cong \bigoplus_{\ell=0}^{n-1} V_\ell.$$

Recover the quantum numbers and transition frequencies, and compare them with hydrogen spectroscopy [34] and Pauli's algebraic derivation [33].

1.8 Monoidal and braided categories

A $\mathbb{C}$ -linear category is a category whose morphism sets

$$\mathrm{Hom}_{\mathcal{C}}(x, y)$$

are complex vector spaces and whose composition is bilinear.

A $\mathbb{C}$ -linear monoidal category is a $\mathbb{C}$ -linear category equipped with a tensor product $\otimes$ that is bilinear on morphisms and is associative with tensor unit 1, up to coherent natural isomorphism.

Let $\mathcal{C}$ be such a category with finite direct sums. An object x is **simple** if $\mathrm{End}(x) \cong \mathbb{C}$, and $\mathcal{C}$ is **semisimple** if every object is a finite direct sum of simple objects. Assume that 1 is simple, that there are finitely many isomorphism classes of simple objects, and that $\mathcal{C}$ is **rigid**: every x has a dual x^* with evaluation and coevaluation maps satisfying the snake identities.

A **pivotal structure** on $\mathcal{C}$ is a monoidal natural isomorphism $j_x : x \rightarrow x^{**}$. It defines the right and left traces of $f : x \rightarrow x$ by

$$\begin{aligned} \mathrm{tr}_R(f) &= \mathrm{ev}_{x^*}(j_x \otimes 1_{x^*})(f \otimes 1_{x^*}) \mathrm{coev}_x, \\ \mathrm{tr}_L(f) &= \mathrm{ev}_x(1_{x^*} \otimes f)(1_{x^*} \otimes j_x^{-1}) \mathrm{coev}_{x^*}. \end{aligned}$$

The pivotal structure is **spherical** if $\mathrm{tr}_L(f) = \mathrm{tr}_R(f)$ for every endomorphism f . Assume that $\mathcal{C}$ has a spherical pivotal structure, write the common trace as $\mathrm{tr}(f)$, and set $d_x = \mathrm{tr}(1_x)$. Assume that the categorical dimensions of nonzero objects are positive real numbers.

For $x, y, z \in \mathcal{C}$, write

$$F^{xyz} : (x \otimes y) \otimes z \longrightarrow x \otimes (y \otimes z)$$

for the associator. A **braiding** is a family of isomorphisms

$$c_{x,y} : x \otimes y \longrightarrow y \otimes x.$$

For $f : x \rightarrow x'$ and $g : y \rightarrow y'$, it satisfies $(g \otimes f)c_{x,y} = c_{x',y'}(f \otimes g)$. The associator satisfies the pentagon identity, and the associator and braiding satisfy the hexagon identities. Suppressing associators, the latter are

$$\begin{aligned} c_{x,y \otimes z} &= (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z), \\ c_{x \otimes y,z} &= (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}). \end{aligned}$$

The **monodromy** of x and y is the double braiding

$$M_{x,y} = c_{y,x}c_{x,y} : x \otimes y \longrightarrow x \otimes y.$$

Thus $c_{x,y}$ is one crossing, while $M_{x,y}$ is a full braid. If $x \otimes y$ is simple, identify $M_{x,y}$ with its scalar in $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Decompositions of tensor products of simple objects are **fusion rules**, and the spaces $\mathrm{Hom}_{\mathcal{C}}(z, x \otimes y)$ are **fusion spaces**.

An object x is **invertible** if $x \otimes x^* \cong 1$.

Exercise 26 (The $\nu = 1/3$ anyonic phase advances by $2\pi/3$)

Assume that every simple object is invertible and that their isomorphism classes form $\mathbb{Z}/m$, generated by a . Classify its monodromy, writing $M_{r,s} = M_{a^r, a^s}$, and show that, for some $k \bmod m$,

$$M_{r,s} = \exp\left(\frac{2\pi i k r s}{m}\right).$$

For $k = 1$ and $m = 3$, derive the $2\pi/3$ phase advance per enclosed anyon and compare it with [50].

Exercise 27 (Reshetikhin–Turaev TQFT)

A ribbon fusion category is a braided fusion category with a dual-compatible twist θ satisfying $\theta_{x \otimes y} = M_{x,y}(\theta_x \otimes \theta_y)$. For simple objects x_i , define $S_{ij} = \text{tr}(M_{x_i, x_j})$ and call the category modular when S is invertible. Prove that every modular ribbon fusion category $\mathcal{C}$ defines a symmetric monoidal functor [44]

$$Z_{\mathcal{C}} : \widehat{\text{Bord}}_3^{\mathcal{C}} \longrightarrow \text{Vect}_{\mathbb{C}},$$

where $\widehat{\text{Bord}}_3^{\mathcal{C}}$ is the anomaly-corrected category of oriented three-dimensional bordisms with $\mathcal{C}$ -colored ribbon graphs.

2 Geometry

2.1 Differential forms and cohomology

A **smooth n -manifold** is a Hausdorff, second-countable topological space equipped with a maximal atlas of charts to open subsets of $\mathbb{R}^n$ whose transition maps are smooth. A smooth map $f : M \rightarrow N$ has a tangent map $df : TM \rightarrow TN$. The **tangent bundle** TM is the union of the tangent spaces, the **cotangent bundle** is T^*M , and a **vector field** is a smooth section of TM .

For an n -dimensional real vector space V , its **determinant line** is $\det(V^*) = \Lambda^n V^*$. An **orientation** of V is a choice of one of the two connected components of $\det(V^*) \setminus \{0\}$; the elements of the chosen component are positive. An orientation of an n -manifold M is a smoothly varying choice of orientation of its tangent spaces.

The exterior powers $\Lambda^k T_p^* M$ are those defined in the Algebra section. The space of **differential n -forms** is $\Omega^n(M) = \Gamma(\Lambda^n T^* M)$. On an oriented n -manifold, a **volume form** is a nowhere-vanishing element of $\Omega^n(M)$ that is positive with respect to the orientation. The exterior derivative $d : \Omega^n(M) \rightarrow \Omega^{n+1}(M)$ extends the differential of smooth functions and satisfies

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$. A **singular n -simplex** is a smooth map $\sigma : \Delta^n \rightarrow M$ where Δ^n is the standard n -simplex. The collection of singular n -simplices in M freely

generates the abelian group $C_n(M)$ of **singular n -chains** in M . The **boundary map** $\partial : C_n(M) \rightarrow C_{n-1}(M)$ is defined on each simplex $\sigma : \Delta^n \rightarrow M$ by

$$\partial\sigma = \sum_{j=0}^n (-1)^j \sigma \circ f_j,$$

where $f_j : \Delta^{n-1} \rightarrow \Delta^n$ is the inclusion of the face opposite the j -th vertex of Δ^n .

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups A_n and homomorphisms $d_n : A_n \rightarrow A_{n-1}$ such that $d_{n-1} \circ d_n = 0$ for all n . The n -th **homology group** H_n of the chain complex is

$$H_n = \ker(d_n) / \operatorname{im}(d_{n+1}).$$

The **singular homology** $H_n(M)$ is the homology of $(C_\bullet(M), \partial)$.

A **cochain complex** $(A^n, d^n)_n$ is a sequence of abelian groups and homomorphisms $d^n : A^n \rightarrow A^{n+1}$ satisfying $d^{n+1}d^n = 0$. Its n th **cohomology group** is

$$H^n(A^\bullet) = \ker d^n / \operatorname{im} d^{n-1}.$$

For an abelian group G , the **singular cochains** are

$$C^n(M; G) = \operatorname{Hom}(C_n(M), G), \quad (d\varphi)(c) = \varphi(\partial c),$$

and their cohomology is the **singular cohomology** $H^n(M; G)$.

The **de Rham cohomology** of M is

$$H_{\text{dR}}^n(M) = \ker(d : \Omega^n(M) \rightarrow \Omega^{n+1}(M)) / \operatorname{im}(d : \Omega^{n-1}(M) \rightarrow \Omega^n(M)).$$

For a smooth map $f : N \rightarrow M$, the **pullback** of $\alpha \in \Omega^p(M)$ is

$$(f^*\alpha)_x(v_1, \dots, v_p) = \alpha_{f(x)}(df_x v_1, \dots, df_x v_p).$$

Exercise 28 (de Rham theorem)

Prove that integration over smooth singular cycles induces a natural isomorphism between de Rham cohomology and singular cohomology with real coefficients.

Exercise 29 (Cohomology produces the Aharonov–Bohm phase)

Outside an idealized solenoid, let

$$M = \mathbb{R}^3 \setminus \{(0, 0, z) : z \in \mathbb{R}\}, \quad d\theta = \frac{-y dx + x dy}{x^2 + y^2}, \quad A = \frac{\Phi}{2\pi} d\theta.$$

For a particle of charge q , use de Rham cohomology to derive the gauge-invariant phase

$$\exp\left(\frac{iq}{\hbar} \int_\gamma A\right)$$

and the Aharonov–Bohm interference shift $q\Phi/\hbar$ [47]. Explain how the shielded-flux electron-holography experiment of Tonomura et al. isolates this holonomy, and compare its measured relative phase shift with the prediction [48].

2.2 Hodge theory and electromagnetism

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric section $g \in \Gamma(T^*M \otimes T^*M)$. On an oriented pseudo-Riemannian n -manifold, the metric induces a bilinear form $\langle \cdot, \cdot \rangle_g$ on differential forms. A **metric volume form** vol_g is a positive volume form satisfying

$$|\langle \text{vol}_g, \text{vol}_g \rangle_g| = 1.$$

The **Hodge star** is an operator $\star : \Omega^p(M) \rightarrow \Omega^{n-p}(M)$ satisfying

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_g \text{vol}_g.$$

Exercise 30 (Metric volume and Hodge star)

Prove that an oriented pseudo-Riemannian manifold has a unique metric volume form and a unique Hodge-star operator. Determine $\star^2$ in terms of the degree and signature of the metric.

On an oriented Lorentzian four-manifold, the electromagnetic field strength is a two-form F and the electric four-current is a three-form J .

Exercise 31 (Maxwell's equations imply charge conservation)

From Maxwell's equations

$$dF = 0, \quad d \star F = J.$$

Derive charge conservation in differential and integral form. On Minkowski spacetime, write

$$F = B - dt \wedge E, \quad J = \rho \text{vol}_3 - dt \wedge \star_3 j,$$

and recover the classical Maxwell equations and continuity equation.

Exercise 32 (Gauge invariance forces charge conservation)

For $F = dA$, consider

$$S[A] = -\frac{1}{2} \int_M F \wedge \star F + \int_M A \wedge J.$$

Derive Maxwell's equations and prove that gauge invariance is equivalent to charge conservation, assuming vanishing boundary terms.

2.3 Vector bundles and connections

A **smooth vector bundle** $p : E \rightarrow M$ is a smooth family of vector spaces locally isomorphic to a product $U \times V \rightarrow U$. For a smooth map $f : N \rightarrow M$, its **pullback bundle** is

$$f^*E = \{(x, e) \in N \times E : f(x) = p(e)\} \longrightarrow N.$$

Write $\Omega^0(M, E) = \Gamma(E)$. A **connection** on E is a ground-field linear map

$$\nabla : \Omega^0(M, E) \longrightarrow \Omega^1(M, E)$$

satisfying $\nabla(fs) = df \otimes s + f\nabla s$. Extend it to E -valued forms by, for $\alpha \in \Omega^p(M)$,

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^p \alpha \wedge \nabla \eta,$$

and define its **curvature** by $F_\nabla = \nabla^2 \in \Omega^2(M, \text{End } E)$. Equip $\text{End } E$ -valued forms with the induced connection.

Exercise 33 (Bianchi identity)

Prove that every vector-bundle connection satisfies $\nabla F_\nabla = 0$.

2.4 K-theory

For a compact Hausdorff space X , let $\text{Vect}(X)$ be the category of finite-rank complex vector bundles with continuous local trivializations over X , equipped with direct sum, tensor product, dual, and pullback. The **Grothendieck group** $K^0(X)$ is generated by isomorphism classes $[E]$ subject to

$$[E \oplus F] = [E] + [F].$$

Define multiplication by $[E][F] = [E \otimes F]$. For a continuous map $f : X \rightarrow Y$ between compact Hausdorff spaces, define pullback by

$$f^* : K^0(Y) \longrightarrow K^0(X), \quad [E] \longmapsto [f^*E].$$

Exercise 34 (Grothendieck group of vector bundles)

*Prove that $(f \circ g)^*E \cong g^*f^*E$ and that pullback preserves direct sums, tensor products, and duals. Let X and Y be compact Hausdorff spaces and let $f : X \rightarrow Y$ be continuous. Prove that direct sum and tensor product induce a commutative ring structure on $K^0(X)$ and that f^* is a ring homomorphism. Prove that*

$$K^0(\text{pt}) \cong \mathbb{Z}$$

via the rank map.

Fix a basepoint $x_0 \in X$, regarded as a map $x_0 : \text{pt} \rightarrow X$. The **reduced K-theory** of the pointed space (X, x_0) is

$$\tilde{K}^0(X) = \ker \left(K^0(X) \xrightarrow{x_0^*} K^0(\text{pt}) \right).$$

For a bundle E of rank r , its **reduced class** is

$$[E] - [\underline{\mathbb{C}}^r],$$

where $\underline{\mathbb{C}}^r = X \times \mathbb{C}^r$ is the trivial rank- r bundle. Write $K^n(X)$ for the generalized cohomology groups extending $K^0(X)$.

Exercise 35 (Stable equivalence and Bott periodicity)

Let E and F be complex vector bundles over a compact Hausdorff space X . Prove that $[E] = [F]$ in $K^0(X)$ if and only if there is a vector bundle G such that

$$E \oplus G \cong F \oplus G.$$

Deduce that the reduced class of E is unchanged after replacing E by $E \oplus \mathbb{C}^n$. Prove Bott periodicity

$$K^{n+2}(X) \cong K^n(X)$$

and deduce that complex K -theory is naturally $\mathbb{Z}/2$ -graded.

For $0 \leq r \leq N$, identify the complex Grassmannian $\text{Gr}_r(\mathbb{C}^N)$ with the space of rank- r orthogonal projections in $M_N(\mathbb{C})$. Its **tautological bundle** has fiber $\text{im } P$ over P . Let BU denote the stable complex Grassmannian obtained from finite Grassmannians by adjoining trivial summands.

Exercise 36 (Stable classification of complex vector bundles)

Classify stable complex vector bundles over a connected finite CW complex X by projection-valued maps, proving

$$K^0(X) \cong [X, \mathbb{Z} \times BU], \quad \tilde{K}^0(X) \cong [X, BU]_*.$$

Exercise 37 (Künneth theorem computes torus K-theory)

For finite CW complexes X and Y with torsion-free K -groups, prove that external product induces the graded isomorphism [1]

$$K^*(X) \hat{\otimes} K^*(Y) \cong K^*(X \times Y).$$

Here $\hat{\otimes}$ denotes graded tensor product. Use Bott periodicity to deduce

$$K^*(\mathbb{T}^d) \cong \Lambda_{\mathbb{Z}}(\eta_1, \dots, \eta_d), \quad |\eta_j| = 1.$$

2.5 Chern classes and the Chern character

Let P be an invariant symmetric k -linear polynomial on the endomorphism algebra of a fiber of E , where invariant means invariant under simultaneous conjugation. Define

$$P(F_{\nabla}) = P(F_{\nabla}, \dots, F_{\nabla}) \in \Omega^{2k}(M).$$

Exercise 38 (Chern–Weil theory produces Chern classes)

Prove that $P(F_{\nabla})$ is closed and that its de Rham class is independent of the connection. Apply this to

$$\det \left(1 + \frac{iF_{\nabla}}{2\pi} \right) = 1 + c_1(E, \nabla) + c_2(E, \nabla) + \dots$$

to define the Chern classes of E [39].

Exercise 39 (Chern character)

Define

$$\text{ch}(E, \nabla) = \text{tr} \exp \left(\frac{iF_\nabla}{2\pi} \right).$$

For compact M , prove that it induces a ring homomorphism

$$\text{ch} : K^0(M) \longrightarrow H_{\text{dR}}^{\text{even}}(M).$$

2.6 Band topology

For the gapped families below, assume that X is compact and connected, so spectra are uniformly bounded and vector-bundle ranks are constant. Let V be a finite-dimensional Hermitian vector space. Exercise 8 supplies the spectral decomposition used below. A continuous family is a map

$$H : X \longrightarrow \text{End}(V)$$

and write $H_x = H(x)$. It is **gapped** if every H_x is self-adjoint and invertible. Given a positively oriented contour Γ enclosing the negative spectra and excluding the positive spectra, define

$$P_x^- = \frac{1}{2\pi i} \int_{\Gamma} (z - H_x)^{-1} dz, \quad E_x^- = \text{im}(P_x^-).$$

Exercise 40 (Gapped families define stable K-theory classes)

Show that the negative spectral subspaces of a gapped family form a vector bundle and that its class in $K^0(X)$ is invariant under gap-preserving homotopy. Show that

$$[E_H^-] - [\mathbb{C}^{\text{rank } E_H^-}] \in \tilde{K}^0(X)$$

is invariant under adjoining constant gapped summands, and that every reduced class is represented by a flattened family $H_x = 1 - 2P_x$.

Exercise 41 (Berry phase as holonomy)

Let $H : X \rightarrow \text{End}(V)$ be a smooth self-adjoint family with a simple isolated eigenvalue λ_x , and let $L \rightarrow X$ be its eigenline bundle. For a local unit eigenvector ψ , define

$$\mathcal{A} = i\langle \psi, d\psi \rangle, \quad \mathcal{F} = d\mathcal{A}.$$

Prove that these local forms define a connection and curvature on L and that the Berry phase around a closed curve γ is its holonomy $\exp(i \int_{\gamma} \mathcal{A})$. For $H_n = n \cdot \sigma$, $n \in S^2$, prove that the phases of the two eigenlines around a loop enclosing oriented solid angle Ω are $\gamma_{\pm} = \mp \Omega/2$ modulo 2π , and compare this prediction with neutron spin rotation [49].

Exercise 42 (K-theory classifies class A band insulators)

A class A band model over a compact connected parameter space X has a smooth gapped Hamiltonian

$$H : X \longrightarrow \text{End}(V)$$

whose negative-energy bundle consists of occupied states; for a crystal, $X = \mathbb{T}^d$. Prove that stable equivalence classes are classified by $\tilde{K}^0(X)$ [4], and compute the strong phases

$$\tilde{K}^0(S^d) \cong \begin{cases} \mathbb{Z}, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

Use Exercise 37 to compute the classifications over $\mathbb{T}^d$ for $d = 1, 2, 3$ and distinguish strong from weak invariants. For $d = 2$, use Exercise 38 to identify the integer invariant with the first Chern number

$$C_1(E_H^-) = \frac{i}{2\pi} \int_{\mathbb{T}^2} \text{tr}(P^- dP^- \wedge dP^-),$$

and prove that a nonzero value obstructs a global smooth frame of occupied states. Assuming the TKNN formula $\sigma_{xy} = C_1(E_H^-)e^2/h$ [6], deduce the quantized Hall resistance and compare its plateaus and geometry independence with the original observation [5].

2.7 Symplectic geometry

For a vector field X , contraction is the map $\iota_X : \Omega^p(M) \rightarrow \Omega^{p-1}(M)$ defined by

$$(\iota_X \alpha)(X_2, \dots, X_p) = \alpha(X, X_2, \dots, X_p).$$

Define the Lie derivative of forms by Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$.

A **symplectic form** is a two-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and $X \mapsto \iota_X \omega$ is a bundle isomorphism $TM \rightarrow T^*M$. Write $C^\infty(M)$ for the smooth real-valued functions on M . For $H \in C^\infty(M)$, define X_H by

$$\iota_{X_H} \omega = dH.$$

Let $\pi : T^*Q \rightarrow Q$ be the cotangent bundle of a configuration manifold. Its **tautological one-form** is

$$\theta_{(q,p)}(\xi) = p(d\pi_{(q,p)}\xi),$$

and its **canonical symplectic form** is $\omega_0 = -d\theta$.

Exercise 43 (The symplectic form produces Hamilton's equations)

For

$$H(q, p) = \frac{1}{2m} \sum_{i=1}^n p_i^2 + V(q),$$

derive Hamilton's equations intrinsically from ω_0 , recover Newton's equation, and solve the harmonic-oscillator flow.

Exercise 44 (Magnetic curvature produces the Lorentz force)

For a particle of charge e in a closed magnetic field $B \in \Omega^2(\mathbb{R}^3)$, derive the Lorentz force from

$$\omega_B = \omega_0 - e\pi^*B \quad \text{and} \quad H(q, p) = \frac{|p|^2}{2m} + e\phi(q),$$

and relate the gauge freedom for $B = dA$ to the Aharonov–Bohm potential.

Exercise 45 (Liouville’s theorem)

Prove that every Hamiltonian flow preserves the symplectic volume $\omega^n/n!$.

Exercise 46 (Phase volume produces ideal-gas thermodynamics)

For Liouville measure $d\mu_L = \omega^n/n!$, show that Hamiltonian evolution preserves the fine-grained Gibbs entropy

$$S_G[\rho] = -k_B \int_M \rho \log \rho \, d\mu_L.$$

For N indistinguishable noninteracting particles in a box, divide phase volume by $N!h^{3N}$ and use

$$H(q, p) = \sum_{a=1}^N \frac{|p_a|^2}{2m}.$$

Derive the Sackur–Tetrode entropy and the ideal-gas laws

$$E = \frac{3}{2}Nk_B T, \quad PV = Nk_B T.$$

Derive the adiabatic and sound-speed laws, including, for $\varrho = mN/V$,

$$c = \sqrt{\frac{5P}{3\varrho}}.$$

Reconcile the entropy increase in free expansion with conservation of fine-grained Gibbs entropy.

For $f, g \in C^\infty(M)$, define

$$\{f, g\} = \omega(X_f, X_g).$$

Exercise 47 (Hamiltonian symmetry produces conservation laws)

Prove that a Hamiltonian f generates a symmetry of H exactly when f is conserved by the flow of H , and distinguish this statement from Noether’s variational theorem [36].

Exercise 48 (Central forces and angular momentum)

On $T^*\mathbb{R}^3$, let

$$H(q, p) = \frac{|p|^2}{2m} + V(|q|), \quad L = q \times p.$$

Derive the angular-momentum Poisson algebra and show that rotational symmetry confines each nonradial trajectory to a fixed plane.

2.8 Riemannian geometry

An **affine connection** is a connection on TM ; write $\nabla_X Y = (\nabla Y)(X)$. For a covector field α , set

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

Define the covariant derivative of general tensor fields by the product rule and by requiring it to commute with contractions.

An affine connection is **torsion-free and metric-compatible** when

$$\nabla_X Y - \nabla_Y X = [X, Y], \quad \nabla g = 0.$$

Such a connection is called a **Levi-Civita connection**.

Exercise 49 (Levi-Civita connection)

Prove that every pseudo-Riemannian manifold admits a unique Levi-Civita connection. Derive the coordinate-free Koszul formula.

An affinely parametrized curve $\gamma : I \rightarrow M$ is a **geodesic** if $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$. The Lie derivative of the metric is

$$(\mathcal{L}_X g)(Y, Z) = X(g(Y, Z)) - g([X, Y], Z) - g(Y, [X, Z]),$$

and X is a **Killing vector field** if $\mathcal{L}_X g = 0$.

Let (M, g) be a pseudo-Riemannian manifold with Levi-Civita connection ∇ .

Exercise 50 (Killing fields and conserved quantities)

Let X be a Killing vector field and let γ be an affinely parametrized geodesic. Prove that

$$\frac{d}{dt} g(\dot{\gamma}, \dot{\gamma}) = 0, \quad \frac{d}{dt} g(X, \dot{\gamma}) = 0.$$

Deduce that $g(X, \dot{\gamma})$ is constant along γ .

The **curvature** of ∇ is the $\text{End}(TM)$ -valued 2-form F_∇ determined by

$$R(X, Y)Z := F_\nabla(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Exercise 51 (Curvature as a tensor)

For $f \in C^\infty(M)$ and vector fields X, Y, Z , prove that

$$R(fX, Y)Z = fR(X, Y)Z, \quad R(X, fY)Z = fR(X, Y)Z,$$

$$R(X, Y)(fZ) = fR(X, Y)Z.$$

Prove that $R(X, Y) = -R(Y, X)$ and deduce that

$$R \in \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes TM).$$

Exercise 52 (Geodesic deviation) Let $\Gamma : (-\varepsilon, \varepsilon) \times I \rightarrow M$ be a smooth variation through affinely parametrized geodesics, and define the vector fields along Γ by

$$T = \frac{\partial \Gamma}{\partial t}, \quad J = \frac{\partial \Gamma}{\partial s}.$$

Prove that $[T, J] = 0$ and that

$$\nabla_T \nabla_T J = R(T, J)T.$$

Exercise 53 (Geodesic deviation produces interferometer strain)

Consider the weak plane-wave metric

$$g = -dt^2 + (1 + h_+(t - z))dx^2 + (1 - h_+(t - z))dy^2 + dz^2, \quad |h_+| \ll 1.$$

Use geodesic deviation to show, to first order, that freely falling mirrors separated along the x - and y -axes acquire opposite fractional length changes $h_+/2$ and $-h_+/2$. Deduce that an equal-arm Michelson interferometer measures differential strain h_+ , and compare this prediction with the peak strain of order 10^{-21} observed for GW150914 [38].

Exercise 54 (Gauss–Bonnet theorem)

Let (M, g) be a compact oriented Riemannian surface with piecewise smooth boundary. With Gaussian curvature K , signed geodesic curvature k_g , and signed exterior angles θ_j , prove [37]

$$\int_M K \operatorname{vol}_g + \int_{\partial M} k_g ds + \sum_j \theta_j = 2\pi \chi(M).$$

Derive its consequences for round spheres, positively curved tori, and closed hyperbolic surfaces.

The trace used below is the ordinary trace defined in the Algebra section. Define the **Ricci tensor**, **scalar curvature**, and **Einstein tensor** by

$$\operatorname{Ric}(X, Y) = \operatorname{tr}(Z \mapsto R(Z, X)Y), \quad S = \operatorname{tr}_g(\operatorname{Ric}), \quad G = \operatorname{Ric} - \frac{1}{2}Sg.$$

For a symmetric covariant two-tensor A , its **divergence** is the covector field

$$(\operatorname{div} A)(Y) = \operatorname{tr}_g((X, Z) \mapsto (\nabla_X A)(Z, Y)).$$

Exercise 55 (Contracted Bianchi identity and conservation)

Assume the second Bianchi identity

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Contract this identity to prove that

$$\operatorname{div} G = 0.$$

Deduce that if T is a symmetric covariant two-tensor satisfying

$$G + \Lambda g = \kappa T$$

for constants Λ, κ with $\kappa \neq 0$, then $\operatorname{div} T = 0$.

2.9 Yang–Mills theory and the Standard Model

Let G be a Lie group with Lie algebra $\mathfrak{g}$. A **principal G -bundle** is a smooth free right G -manifold P with quotient $M = P/G$ such that the projection $P \rightarrow M$ is a locally trivial fiber bundle. For a finite-dimensional representation $\rho : G \rightarrow \mathrm{GL}(V)$, the **associated bundle** is

$$E = P \times_G V = (P \times V) / \sim, \quad (p \cdot g, v) \sim (p, \rho(g)^{-1}v).$$

A **principal connection** is a $\mathfrak{g}$ -valued one-form A on P that is G -equivariant and reproduces the generators of the fundamental vector fields. In a local trivialization it is represented by an ordinary form $A \in \Omega^1(U, \mathfrak{g})$, with curvature

$$F_A = dA + A \wedge A \in \Omega^2(U, \mathfrak{g}).$$

Exercise 56 (Isospin motivates Yang–Mills connections)

Treat the proton and neutron as a nucleon doublet carrying the defining representation of $\mathrm{SU}(2)$. Gauge this symmetry and derive

$$A \mapsto A^g = g^{-1}Ag + g^{-1}dg.$$

Show that the resulting connection differentiates associated-bundle fields and has covariant curvature, and relate this construction to [40].

Exercise 57 (Yang–Mills action)

On an oriented Riemannian four-manifold, consider the Yang–Mills action

$$S(A) = \int_M \mathrm{Tr}(F_A \wedge \star F_A).$$

Prove gauge invariance, derive

$$d_A \star F_A = 0,$$

and recover source-free Maxwell theory in the abelian case.

Exercise 58 (Standard Model symmetry breaking)

Take the Standard Model gauge group to be

$$G_{\mathrm{SM}} = \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y.$$

Write $(\mathbf{n}, \mathbf{m})_Y$ for the representation of dimension n under $\mathrm{SU}(3)_C$, dimension m under $\mathrm{SU}(2)_L$, and weak hypercharge Y . Take one generation of Weyl fermion multiplets to be

$$\begin{aligned} Q_L &= (\mathbf{3}, \mathbf{2})_{1/6}, & u_R &= (\mathbf{3}, \mathbf{1})_{2/3}, & d_R &= (\mathbf{3}, \mathbf{1})_{-1/3}, \\ L_L &= (\mathbf{1}, \mathbf{2})_{-1/2}, & e_R &= (\mathbf{1}, \mathbf{1})_{-1}, \end{aligned}$$

and define $Q = T_3 + Y$. Let ϕ be a section of the Higgs bundle associated to $(\mathbf{1}, \mathbf{2})_{1/2}$, with

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \quad D\phi = d\phi + \frac{i}{2}g W^a \sigma_a \phi + \frac{i}{2}g' B\phi,$$

where $\mu^2, \lambda > 0$ and g, g' are coupling constants. From these data derive the fermion charges, the symmetry breaking $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$, the W and Z masses, the massless photon, and cancellation of the hypercharge anomalies. Identify the $W^\pm \rightarrow e^\pm \nu$ missing-transverse-energy and $Z^0 \rightarrow \ell^+ \ell^-$ invariant-mass signatures permitted by these representations, and compare them with the original UA1 observations [41, 42].

2.10 Chern–Simons theory and TQFT

Exercise 59 (Chern–Simons transgression)

For two connections on E , set

$$\nabla^t = (1-t)\nabla^0 + t\nabla^1, \quad A = \nabla^1 - \nabla^0, \quad F_t = F_{\nabla^t},$$

where $A \in \Omega^1(M, \text{End } E)$, and define

$$\text{CS}_P(\nabla^0, \nabla^1) = k \int_0^1 P(A, F_t, \dots, F_t) dt.$$

Prove that

$$P(F_{\nabla^1}) - P(F_{\nabla^0}) = d \text{CS}_P(\nabla^0, \nabla^1).$$

Let $\mathfrak{g} \subseteq M_N(\mathbb{C})$ be a matrix Lie algebra whose trace pairing is nondegenerate. For $A \in \Omega^1(M, \mathfrak{g})$, wedge products include matrix multiplication. As above, write

$$F_A = dA + A \wedge A$$

and define

$$\text{CS}(A) = \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

Exercise 60 (Chern–Simons form)

Prove that

$$d \text{CS}(A) = \text{tr}(F_A \wedge F_A).$$

For a closed oriented three-manifold M , define

$$S_{\text{CS}}(A) = \int_M \text{CS}(A).$$

Exercise 61 (Chern–Simons equation)

For $\alpha \in \Omega^1(M, \mathfrak{g})$, set $A_t = A + t\alpha$. Prove that

$$\left. \frac{d}{dt} \right|_{t=0} S_{\text{CS}}(A_t) = 2 \int_M \text{tr}(\alpha \wedge F_A).$$

Deduce that the critical points satisfy $F_A = 0$.

For $k \in \mathbb{R}$, set $S_k = (k/4\pi)S_{\text{CS}}$.

Exercise 62 (Gauge transformation)

Let G be a compact, connected, simple, and simply connected matrix Lie group with Lie algebra $\mathfrak{g}$. For a smooth map $g : M \rightarrow G$, define

$$A^g = g^{-1}Ag + g^{-1}dg.$$

Normalize the invariant trace so that

$$\nu(g) = \frac{1}{24\pi^2} \int_M \text{tr}((g^{-1}dg)^3) \in \mathbb{Z}.$$

Prove that

$$\text{CS}(A^g) - \text{CS}(A) = -d \text{tr}(dg g^{-1} \wedge A) - \frac{1}{3} \text{tr}((g^{-1}dg)^3).$$

Deduce that $\exp(iS_k)$ is invariant under all gauge transformations when $k \in \mathbb{Z}$.

Now restrict to $2 + 1$ spacetime dimensions. Let M be a closed oriented three-manifold and let $\ell > 0$, so the cosmological constant is $\Lambda = -\ell^{-2}$. After identifying the coframe representation with $\mathfrak{sl}(2, \mathbb{R})$, let

$$e, \omega \in \Omega^1(M, \mathfrak{sl}(2, \mathbb{R})), \quad A^\pm = \omega \pm \frac{1}{\ell}e.$$

The pair (A^+, A^-) has gauge Lie algebra $\mathfrak{sl}(2, \mathbb{R}) \oplus \mathfrak{sl}(2, \mathbb{R})$. Write

$$d_\omega e = de + \omega \wedge e + e \wedge \omega, \quad F_\omega = d\omega + \omega \wedge \omega.$$

Exercise 63 (Flat connections encode three-dimensional gravity)

Prove that $F_{A^+} = F_{A^-} = 0$ is equivalent to

$$d_\omega e = 0, \quad F_\omega + \frac{1}{\ell^2}e \wedge e = 0.$$

Assuming that $e : TM \rightarrow \mathfrak{sl}(2, \mathbb{R})$ is fiberwise invertible, identify these equations with the torsion-free condition and the Einstein equation with constant negative curvature.

Exercise 64 (Three-dimensional gravity is a Chern–Simons theory)

For the first-order Einstein–Hilbert functional

$$S_{\text{EH}}(e, \omega) = \int_M \text{tr} \left(e \wedge F_\omega + \frac{1}{3\ell^2} e \wedge e \wedge e \right).$$

Prove that, up to normalization and an exact term, it equals

$$S_{\text{CS}}(A^+) - S_{\text{CS}}(A^-)$$

[45, 46].

For a connection A and a finite-dimensional representation V of its gauge group, parallel transport around an oriented closed curve γ gives the holonomy $\text{Hol}_\gamma(A)$. Define the **Wilson loop**

$$W_V(\gamma, A) = \text{tr}_V(\text{Hol}_\gamma(A)).$$

For a framed oriented link $L = \gamma_1 \sqcup \cdots \sqcup \gamma_r$ labeled by $V_1, \dots, V_r$, set

$$W_L(A) = \prod_{j=1}^r W_{V_j}(\gamma_j, A).$$

For a framed link, define the formal Chern–Simons expectation value [43]

$$\langle W_L \rangle_k = \frac{\int e^{iS_k(A)} W_L(A) \mathcal{D}A}{\int e^{iS_k(A)} \mathcal{D}A}.$$

Exercise 65 (Wilson loops and the Jones polynomial)

For normalized $SU(2)$ theory at level k , set

$$q = \exp\left(\frac{2\pi i}{k+2}\right).$$

Assume the normalized Wilson-loop expectations satisfy

$$q^{-1} \langle W_{L_+} \rangle_k - q \langle W_{L_-} \rangle_k = \left(q^{1/2} - q^{-1/2}\right) \langle W_{L_0} \rangle_k$$

and $\langle W_\circ \rangle_k = 1$. Show that they determine the Jones polynomial, account for framing dependence, and agree with the Reshetikhin–Turaev invariant of Exercise 27.

3 Analysis

3.1 Operators on Hilbert space

A **Hilbert space** is a complete complex inner-product space. Write $\mathcal{B}(H, K)$ for the bounded operators $H \rightarrow K$ with

$$\|T\| = \sup_{\|x\|=1} \|Tx\|, \quad \mathcal{B}(H) = \mathcal{B}(H, H).$$

A **C^* -algebra** is a complex Banach $*$ -algebra A satisfying $\|a^*a\| = \|a\|^2$. An element $a \in A$ is **positive**, written $a \geq 0$, if $a = b^*b$ for some $b \in A$. The algebra $M_n(A)$ has its standard C^* -algebra structure.

Exercise 66 (Hilbert-space operators form a C^* -algebra)

Prove the Cauchy–Schwarz inequality, construct the adjoint of every $T \in \mathcal{B}(H, K)$, and prove

$$(ST)^* = T^*S^*, \quad \|T^*T\| = \|T\|^2.$$

Deduce that $\mathcal{B}(H)$ is a C^* -algebra and that $M_n(\mathcal{B}(H)) \cong \mathcal{B}(H^{\oplus n})$ isometrically as a $*$ -algebra.

3.2 Operator K-theory

For a unital C^* -algebra A , let $M_\infty(A) = \bigcup_{n \geq 1} M_n(A)$ under $x \mapsto x \oplus 0$. Projections $p, q \in M_\infty(A)$ are **stably Murray–von Neumann equivalent** if, after adjoining zero blocks, there is $v \in M_\infty(A)$ with

$$v^*v = p, \quad vv^* = q.$$

Let $V(A)$ be the resulting set of equivalence classes with addition induced by direct sum. Define $K_0(A)$ as the Grothendieck group of $V(A)$.

Let $U_\infty(A) = \bigcup_{n \geq 1} U(M_n(A))$ under $u \mapsto u \oplus 1$, and define $K_1(A) = \pi_0(U_\infty(A))$. For a nonunital C^* -algebra A , let $\tilde{A}$ be its unitization with quotient $\epsilon : \tilde{A} \rightarrow \mathbb{C}$, and define

$$K_i(A) = \ker \left(K_i(\tilde{A}) \xrightarrow{\epsilon_*} K_i(\mathbb{C}) \right), \quad i = 0, 1.$$

Exercise 67 (Stable projections and unitaries define K-groups)

Prove that direct sum makes $V(A)$ a commutative monoid, that $K_1(A)$ is an abelian group, and that K_0 and K_1 are functorial for $$ -homomorphisms. Prove the stable isomorphisms*

$$K_i(M_n(A)) \cong K_i(A)$$

and compute

$$K_0(\mathbb{C}) \cong \mathbb{Z}, \quad K_1(\mathbb{C}) = 0.$$

An operator $K \in \mathcal{B}(H, K')$ is **compact** if the image of its closed unit ball has compact closure. Write $\mathcal{K}(H)$ for the compact operators on H .

Exercise 68 (Compact operators carry integer-valued K-theory)

For an infinite-dimensional separable Hilbert space H , prove that $\mathcal{K}(H)$ is the norm-closed ideal generated by finite-rank operators. Classify projections in $M_\infty(\mathcal{K}(H))$ by rank and prove

$$K_0(\mathcal{K}(H)) \cong \mathbb{Z}, \quad K_1(\mathcal{K}(H)) = 0.$$

For compact Hausdorff X , write $C(X)$ for the unital C^* -algebra of continuous complex-valued functions on X with pointwise operations.

Exercise 69 (Operator K-theory recovers vector bundles)

For compact Hausdorff X , associate to $p \in M_n(C(X))$ the bundle with fiber $\text{im } p(x)$. Prove the Serre–Swan equivalence between complex vector bundles over X and finitely generated projective $C(X)$ -modules, where projective means isomorphic to a direct summand of $C(X)^n$ [8]. Deduce the natural isomorphism

$$K_0(C(X)) \cong K^0(X).$$

Identify this description with the projection-valued classification of Exercise 36.

3.3 Fredholm operators and index theory

An operator $T : H \rightarrow K'$ is **Fredholm** if its kernel and cokernel are finite dimensional and its image is closed; define

$$\text{ind}(T) = \dim \ker T - \dim \text{coker } T.$$

Exercise 70 (The Calkin boundary map is the Fredholm index)

Let H be infinite-dimensional and separable and set $\mathcal{Q}(H) = \mathcal{B}(H)/\mathcal{K}(H)$. Prove that $T \in \mathcal{B}(H)$ is Fredholm exactly when its image $q(T) \in \mathcal{Q}(H)$ is invertible. Construct the six-term K-theory sequence of [7]

$$0 \longrightarrow \mathcal{K}(H) \longrightarrow \mathcal{B}(H) \xrightarrow{q} \mathcal{Q}(H) \longrightarrow 0$$

Compute $K_i(\mathcal{B}(H)) = 0$, use polar retraction to regard $q(T)$ as a K_1 -class, and prove, with the index convention for the boundary map,

$$\partial : K_1(\mathcal{Q}(H)) \xrightarrow{\cong} K_0(\mathcal{K}(H)) \cong \mathbb{Z}, \quad \partial[q(T)] = [P_{\ker T}] - [P_{\ker T^*}] = \text{ind}(T).$$

Deduce that the Fredholm index is locally constant, additive under composition, and invariant under compact perturbations.

Let $\text{Fred}(H)$ denote the Fredholm operators on H , with the operator-norm topology.

Exercise 71 (Fredholm families represent K-theory)

For an infinite-dimensional separable Hilbert space H and compact Hausdorff space X , use Exercise 70 to construct the index bundle of a norm-continuous family $X \rightarrow \text{Fred}(H)$ and prove the Atiyah–Jänich isomorphism [1]

$$[X, \text{Fred}(H)] \cong K^0(X).$$

Let M be a closed smooth manifold and let $E, F \rightarrow M$ be complex vector bundles. An order- m **differential operator** $D : \Gamma(E) \rightarrow \Gamma(F)$ is locally of the form $D = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha$. Its **principal symbol** is

$$\sigma_D(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha : E_x \longrightarrow F_x.$$

The operator is **elliptic** if $\sigma_D(x, \xi)$ is invertible for $\xi \neq 0$. Write $K_c^0(Y) = \tilde{K}^0(Y^+)$ for compactly supported K-theory. The **symbol class** of an elliptic operator is

$$[\sigma_D] = [\pi^* E, \pi^* F, \sigma_D] \in K_c^0(T^*M).$$

Define $\text{ind}_{\text{top}} : K_c^0(T^*M) \rightarrow \mathbb{Z}$ by the K-theoretic pushforward to a point.

Exercise 72 (Atiyah–Singer index theorem)

Prove that an elliptic operator D has a Fredholm extension and that [2]

$$\text{ind}(D) = \dim \ker D - \dim \ker D^* = \text{ind}_{\text{top}}[\sigma_D].$$

For a Riemannian metric on M , apply this to the de Rham operator $d + d^* : \Omega^{\text{even}}(M) \rightarrow \Omega^{\text{odd}}(M)$ and recover $\text{ind}(d + d^*) = \chi(M)$.

Exercise 73 (Instanton charge produces the chiral anomaly)

Let A be a finite-action anti-self-dual $SU(N)$ connection on $\mathbb{R}^4$ extending to a Hermitian bundle $E \rightarrow S^4$. For skew-Hermitian curvature F_A and the fundamental trace, define its topological charge by

$$Q(A) = \frac{1}{8\pi^2} \int_{S^4} \text{tr}(F_A \wedge F_A) = \langle c_2(E), [S^4] \rangle.$$

Prove that $Q(A) \in \mathbb{Z}$. Let $S^\pm$ be the half-spinor bundles and $D_A^\pm$ the coupled chiral Dirac operators. Use Exercise 72 to prove

$$\dim \ker D_A^- - \dim \ker D_A^+ = Q(A).$$

For N_f massless Dirac fermions in the fundamental representation, deduce the integrated chiral anomaly [3]

$$\Delta Q_5 = \int_{\mathbb{R}^4} d(\star J_5) = 2N_f Q(A).$$

3.4 Operator spaces and completely bounded maps

A **concrete operator space** is a closed linear subspace $E \subseteq \mathcal{B}(H, K)$. Its n th matrix level is

$$M_n(E) \subseteq \mathcal{B}(H^{\oplus n}, K^{\oplus n})$$

with the inherited operator norm. For a linear map $\phi : E \rightarrow F$, define

$$\phi_n([x_{ij}]) = [\phi(x_{ij})], \quad \|\phi\|_{\text{cb}} = \sup_{n \geq 1} \|\phi_n\|.$$

The map is a **complete contraction**, **complete isometry**, or **completely bounded** when every ϕ_n is contractive, isometric, or uniformly bounded, respectively. Write $\text{CB}(E, F)$ for the completely bounded maps.

Exercise 74 (Ruan's axioms characterize concrete operator spaces)

Show that concrete matrix norms satisfy

$$\|x \oplus y\| = \max\{\|x\|, \|y\|\}, \quad \|\alpha x \beta\| \leq \|\alpha\| \|x\| \|\beta\|.$$

Conversely, prove that a vector space with matrix norms satisfying these axioms admits a completely isometric embedding into $\mathcal{B}(H)$ [9].

The **row** and **column operator spaces** are

$$R_n = M_{1,n} \subseteq \mathcal{B}(\mathbb{C}^n, \mathbb{C}), \quad C_n = M_{n,1} \subseteq \mathcal{B}(\mathbb{C}, \mathbb{C}^n).$$

Exercise 75 (Matrix levels detect noncommutativity)

For the identity maps between the underlying Hilbert spaces of R_n and C_n , prove

$$\|1 : R_n \rightarrow C_n\|_{\text{cb}} = \|1 : C_n \rightarrow R_n\|_{\text{cb}} = \sqrt{n}.$$

For transposition $t : M_n \rightarrow M_n$, prove

$$\|t\| = 1, \quad \|t\|_{\text{cb}} = n.$$

Explain why Banach-space norm alone forgets both distinctions.

Exercise 76 (Smith's lemma stabilizes maps into matrices)

For an operator space E and a linear map $\phi : E \rightarrow M_n$, prove [10]

$$\|\phi\|_{\text{cb}} = \|\phi_n\|.$$

Deduce that complete boundedness is a finite computation whenever the codomain is a fixed matrix algebra.

3.5 Operator-space duality and Haagerup tensoring

The **operator-space dual** of E is the Banach dual E^* with matrix norms determined by

$$M_n(E^*) = \text{CB}(E, M_n)$$

isometrically. Rectangular spaces $M_{p,q}(E)$ carry the norms inherited from $\mathcal{B}(H^{\oplus q}, K^{\oplus p})$. For $u = \sum_{k=1}^r x_k \otimes y_k \in E \otimes F$, define

$$\|u\|_h = \inf \left\{ \|[x_1 \ \cdots \ x_r]\|_{M_{1,r}(E)} \|[y_1 \ \cdots \ y_r]^T\|_{M_{r,1}(F)} \right\}.$$

The completion is the **Haagerup tensor product** $E \otimes_h F$.

Exercise 77 (Rows and columns give Schatten norms)

Under $e_i \otimes e_j \mapsto E_{ij}$, prove the complete isometries

$$C_n \otimes_h R_n \cong M_n, \quad R_n \otimes_h C_n \cong S_1^n,$$

where M_n has operator norm and S_1^n has trace norm. Interpret the reversal of the tensor factors as the distinction between largest singular value and total singular mass.

Exercise 78 (Trace norm governs state discrimination)

Let $\rho_0, \rho_1 \in M_n$ be positive with trace one, and let p_0, p_1 be probabilities with $p_0 + p_1 = 1$. A **binary measurement** is a pair $M_0, M_1 \geq 0$ with $M_0 + M_1 = 1$. Define $\|x\|_1 = \text{tr} \sqrt{x^* x}$. For

$$p_{\text{succ}} = p_0 \text{tr}(M_0 \rho_0) + p_1 \text{tr}(M_1 \rho_1),$$

prove the Helstrom formula [12]

$$\max_{(M_0, M_1)} p_{\text{succ}} = \frac{1}{2} (1 + \|p_0 \rho_0 - p_1 \rho_1\|_1).$$

Determine an optimal measurement from the positive spectral subspace of $p_0 \rho_0 - p_1 \rho_1$, and use $R_n \otimes_h C_n \cong S_1^n$ to interpret distinguishability as a Haagerup tensor norm.

For a bilinear map $u : E \times F \rightarrow G$, define

$$u_n(x, y)_{ij} = \sum_{k=1}^n u(x_{ik}, y_{kj}), \quad \|u\|_{\text{mb}} = \sup_n \|u_n\|.$$

Such a map is **multiplicatively bounded** if this supremum is finite.

Exercise 79 (Haagerup tensoring linearizes products)

Prove that u is multiplicatively bounded exactly when its linearization extends to a completely bounded map

$$\tilde{u} : E \otimes_h F \longrightarrow G, \quad \tilde{u}(x \otimes y) = u(x, y),$$

and prove $\|\tilde{u}\|_{\text{cb}} = \|u\|_{\text{mb}}$. Apply this to multiplication in a C^ -algebra.*

3.6 Operator systems and completely positive maps

An **operator system** is a self-adjoint unital subspace $S \subseteq \mathcal{B}(H)$. A linear map $\Phi : S \rightarrow T$ is **positive** if $\Phi(S_+) \subseteq T_+$, and **completely positive** if every $\Phi_n : M_n(S) \rightarrow M_n(T)$ is positive.

Exercise 80 (Partial transpose detects entanglement)

A state in $M_m \otimes M_n$ is a positive element ρ with $\text{tr } \rho = 1$. Call it separable if $\rho = \sum_k p_k \sigma_k \otimes \tau_k$ for states σ_k, τ_k and probabilities p_k . Prove that transposition $t : M_n \rightarrow M_n$ is positive and that every separable state has positive partial transpose [13]

$$\rho^\Gamma = (1 \otimes t)(\rho) \geq 0.$$

For $\Omega_2 = (|00\rangle + |11\rangle)/\sqrt{2}$, compute the spectrum of $(|\Omega_2\rangle\langle\Omega_2|)^\Gamma$ and deduce that t is not 2-positive. Assume the Peres–Horodecki theorem that positivity of the partial transpose is also sufficient for separability in 2×2 and 2×3 systems [14]. Apply it to

$$\rho_p = p|\psi^-\rangle\langle\psi^-| + (1-p)\frac{1_4}{4}, \quad |\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},$$

and prove that ρ_p is separable exactly when $p \leq 1/3$.

For $\Phi : M_n \rightarrow M_m$, its **Choi matrix** is

$$C_\Phi = \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij}) \in M_n \otimes M_m.$$

Exercise 81 (Choi matrices classify completely positive maps)

Prove that the following are equivalent [11]:

$$\Phi \text{ is completely positive,} \quad C_\Phi \geq 0, \quad \Phi(x) = \sum_{\alpha} V_{\alpha} x V_{\alpha}^*.$$

Determine the least number of Kraus operators from $\text{rank } C_\Phi$ and express trace preservation and unitality as partial-trace conditions on C_Φ .

Exercise 82 (Choi states enable process tomography)

Let $\Omega_n = n^{-1/2} \sum_{i=1}^n e_i \otimes e_i$ and let $\Phi : M_n \rightarrow M_m$ be completely positive and trace preserving. Define

$$\rho_\Phi = (1 \otimes \Phi)(|\Omega_n\rangle\langle\Omega_n|).$$

Prove

$$\rho_\Phi = \frac{1}{n} C_\Phi, \quad \Phi(x) = n \operatorname{Tr}_1((x^\top \otimes 1)\rho_\Phi), \quad \operatorname{Tr}_2 \rho_\Phi = \frac{1}{n} 1.$$

Here Tr_1 and Tr_2 are the partial traces over the first and second tensor factors. Call a bipartite probe $\sigma \in M_n \otimes M_n$ full operator-Schmidt rank if $\sigma = \sum_{k=1}^{n^2} A_k \otimes B_k$ with both families linearly independent. Prove that $(1 \otimes \Phi)(\sigma)$ determines every linear map $\Phi : M_n \rightarrow M_m$ exactly when σ has full operator-Schmidt rank. Construct a separable probe with this property, and compare the entangled- and separable-probe process reconstructions reported in the photonic experiment of Altepeter et al. [15].

Exercise 83 (Stinespring dilation turns positivity into geometry)

For a completely positive map $\Phi : A \rightarrow \mathcal{B}(H)$ from a unital C^* -algebra, construct a $*$ -representation $\pi : A \rightarrow \mathcal{B}(K)$ and $V : H \rightarrow K$ such that [32]

$$\Phi(a) = V^* \pi(a) V, \quad V^* V = \Phi(1).$$

Prove uniqueness of a minimal dilation up to unitary equivalence.

For $E \subseteq \mathcal{B}(H, K)$, define its **Paulsen operator system**

$$\mathcal{S}(E) = \left\{ \begin{pmatrix} \lambda 1_K & x \\ y^* & \mu 1_H \end{pmatrix} : \lambda, \mu \in \mathbb{C}, x, y \in E \right\} \subseteq \mathcal{B}(K \oplus H).$$

Exercise 84 (The 2×2 trick converts contractions into positivity)

For $\phi : E \rightarrow \mathcal{B}(H_1, H_2)$, prove that ϕ is a complete contraction exactly when the unital map

$$\begin{pmatrix} \lambda 1_K & x \\ y^* & \mu 1_H \end{pmatrix} \mapsto \begin{pmatrix} \lambda 1_{H_2} & \phi(x) \\ \phi(y)^* & \mu 1_{H_1} \end{pmatrix}$$

is completely positive. Use this operator-system encoding to derive the completely contractive extension theorem.

Exercise 85 (Wittstock factorizes completely bounded maps)

For a completely bounded map $\phi : A \rightarrow \mathcal{B}(H)$, prove that there are a representation $\pi : A \rightarrow \mathcal{B}(K)$ and operators $V, W : H \rightarrow K$ such that [16]

$$\phi(a) = V^* \pi(a) W, \quad \|\phi\|_{\text{cb}} = \inf \{ \|V\| \|W\| : \phi(\cdot) = V^* \pi(\cdot) W \}.$$

Deduce that every completely bounded map is a linear combination of completely positive maps.

3.7 Quantum channels and matrix-level distinguishability

For $x \in M_n$, write $\|x\|_1 = \text{tr} \sqrt{x^*x}$. A **state** is $\rho \in M_n$ with $\rho \geq 0$ and $\text{tr} \rho = 1$. The **partial trace** $\text{Tr}_B : M_n \otimes M_m \rightarrow M_n$ is characterized by

$$\text{tr}(a \text{Tr}_B x) = \text{tr}((a \otimes 1)x).$$

A **Schrödinger-picture quantum channel** is a completely positive trace-preserving map $\Phi : M_n \rightarrow M_m$. Its **Heisenberg adjoint** $\Phi^* : M_m \rightarrow M_n$ is determined by

$$\text{tr}(\Phi(\rho)a) = \text{tr}(\rho\Phi^*(a)).$$

Exercise 86 (Schmidt spectra measure entanglement)

For $\psi \in H_A \otimes H_B$, apply singular-value decomposition to $T_\psi : H_A^* \rightarrow H_B$, where $T_\psi(f) = (f \otimes 1)(\psi)$, and obtain

$$\psi = \sum_{i=1}^r s_i a_i \otimes b_i.$$

Characterize product vectors by $r = 1$, identify s_i^2 as the nonzero eigenvalues of both reduced states, and compute

$$S(\psi) = - \sum_i s_i^2 \log s_i^2.$$

Exercise 87 (Channels are isometries with an environment)

Combine Choi's theorem and Stinespring dilation to prove that every channel has equivalent forms

$$\Phi(\rho) = \sum_{\alpha} K_{\alpha} \rho K_{\alpha}^* = \text{Tr}_E(V \rho V^*), \quad V^* V = 1.$$

Describe the Choi–Jamiołkowski correspondence and determine which bipartite states correspond to channels.

For a linear map $\Psi : M_n \rightarrow M_m$, define its **diamond norm** by

$$\|\Psi\|_{\diamond} = \sup_{k \geq 1} \|\Psi \otimes 1_{M_k}\|_{1 \rightarrow 1}.$$

Exercise 88 (The diamond norm is the operational cb norm)

Prove the stabilization and duality identities [17]

$$\|\Psi\|_{\diamond} = \|\Psi \otimes 1_{M_n}\|_{1 \rightarrow 1}, \quad \|\Psi\|_{\diamond} = \|\Psi^*\|_{\text{cb}}.$$

For equally likely channels Φ_0, Φ_1 , prove that the optimal one-use discrimination probability, allowing an entangled ancilla, is

$$p_{\text{succ}} = \frac{1}{2} + \frac{1}{4} \|\Phi_0 - \Phi_1\|_{\diamond}.$$

Exercise 89 (Entanglement improves process discrimination)

Assume the Piani–Watrous theorem that every entangled bipartite state improves the success probability in some channel-discrimination task [18]. Explain why this is an operational manifestation of matrix amplification. Reconstruct the subchannels and unsteerable benchmark in the photonic experiment of Sun et al., and compare its measured steering-assisted success probabilities with that benchmark [19].

Exercise 90 (No channel universally clones pure states)

Suppose a channel C and fixed blank state σ satisfy $C(\rho \otimes \sigma) = \rho \otimes \rho$ for every pure state ρ . Use contractivity of trace distance, or monotonicity of fidelity, to obtain a contradiction from two nonorthogonal states. Recover the unitary inner-product argument of Wootters and Zurek [31].

3.8 Nonlocal correlations and Grothendieck inequalities**Exercise 91 (Every entangled pure two-qubit state violates CHSH)**

Let A_0, A_1 and B_0, B_1 be self-adjoint operators with eigenvalues in $\{-1, 1\}$, and define the CHSH operator

$$C = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Establish the classical CHSH bound 2. For the two-qubit state

$$\psi_\theta = \cos \theta |00\rangle + \sin \theta |11\rangle, \quad 0 \leq \theta \leq \frac{\pi}{4},$$

derive the maximal CHSH value [27]

$$2\sqrt{1 + \sin^2(2\theta)}.$$

Conclude that every entangled pure two-qubit state violates CHSH. Relate the four observables to spacelike-separated measurement settings and compare the quantum violation with a loophole-free Bell experiment using electron spins separated by 1.3 kilometers [28].

A bipartite binary quantum correlation matrix has entries

$$c_{ij} = \langle \psi, (A_i \otimes B_j) \psi \rangle,$$

where ψ is a unit vector. The operators A_i and B_j are self-adjoint unitaries on finite-dimensional Hilbert spaces.

For a real matrix $M = (M_{ij})$, define

$$\beta_C(M) = \max_{\varepsilon_i, \delta_j \in \{-1, 1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|$$

and

$$\beta_Q(M) = \sup_c \left| \sum_{i,j} M_{ij} c_{ij} \right|,$$

where the supremum is over bipartite binary quantum correlation matrices.

Exercise 92 (Tsirelson turns quantum correlations into vectors)

Prove that every binary quantum correlation has a representation [29]

$$c_{ij} = \langle x_i, y_j \rangle_{\mathbb{R}}, \quad \|x_i\|, \|y_j\| \leq 1.$$

Prove the converse using finite-dimensional Clifford representations and deduce the sharp CHSH value $2\sqrt{2}$.

The **real Grothendieck constant** $K_G^{\mathbb{R}}$ is the least constant for which

$$\beta_Q(M) \leq K_G^{\mathbb{R}} \beta_C(M)$$

holds for every real matrix M .

Exercise 93 (Grothendieck bounds quantum advantage uniformly)

Prove that $K_G^{\mathbb{R}} < \infty$ [30]. Use

$$M_{\text{CHSH}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

to prove $K_G^{\mathbb{R}} \geq \sqrt{2}$ and interpret the inequality as a dimension-independent comparison of scalar and Hilbert-space strategies.

For a bilinear form $u : A \times B \rightarrow \mathbb{C}$, define $T_u : A \rightarrow B^*$ by $T_u(a)(b) = u(a, b)$. The form is **jointly completely bounded** when T_u is completely bounded, with $\|u\|_{\text{jcb}} = \|T_u\|_{\text{cb}}$. A **state** on a unital C^* -algebra is a positive functional of norm one.

Exercise 94 (The operator-space Grothendieck inequality)

Prove that for every jointly completely bounded $u : A \times B \rightarrow \mathbb{C}$ there are states f_1, f_2 on A and g_1, g_2 on B such that [20]

$$|u(a, b)| \leq \|u\|_{\text{jcb}} \left(f_1(aa^*)^{1/2} g_1(b^*b)^{1/2} + f_2(a^*a)^{1/2} g_2(bb^*)^{1/2} \right).$$

Deduce a factorization of T_u through a direct sum of row and column Hilbert operator spaces, and recover the scalar Grothendieck pattern on commutative subalgebras.

3.9 Dual operator spaces and von Neumann algebras

For $\mathcal{S} \subseteq \mathcal{B}(H)$, its **commutant** is

$$\mathcal{S}' = \{T \in \mathcal{B}(H) : Tx = xT \text{ for every } x \in \mathcal{S}\}.$$

A **von Neumann algebra** is a unital $*$ -subalgebra $M \subseteq \mathcal{B}(H)$ satisfying $M = M''$. A net (T_λ) converges to T in the **strong operator topology** if $T_\lambda \xi \rightarrow T\xi$ for every $\xi \in H$, and in the **weak operator topology** if

$$\langle T_\lambda \xi, \eta \rangle \rightarrow \langle T\xi, \eta \rangle$$

for every $\xi, \eta \in H$.

Exercise 95 (Bicommutants are dual operator algebras)

Let $A \subseteq \mathcal{B}(H)$ be a unital $*$ -subalgebra. Prove that

$$A'' = \overline{A}^{\text{SOT}} = \overline{A}^{\text{WOT}}.$$

Identify the canonical predual of A'' as a quotient of the trace-class operators.

An operator space E is a **dual operator space** with predual E_* if $E = E_*^*$ completely isometrically. A map between dual operator spaces is **normal** if it is weak- $*$ continuous. The **center** of a von Neumann algebra is $Z(M) = M \cap M'$, and M is a **factor** if $Z(M) = \mathbb{C}1$. A **normal channel** is a normal unital completely positive map.

Exercise 96 (Normal CP maps have Stinespring dilations)

Let M be a von Neumann algebra and let $\Phi : M \rightarrow \mathcal{B}(H)$ be normal and completely positive. Prove Stinespring's theorem [32]: there are a Hilbert space K , a normal representation $\pi : M \rightarrow \mathcal{B}(K)$, and $V : H \rightarrow K$ such that

$$\Phi(x) = V^* \pi(x) V, \quad V^* V = \Phi(1).$$

Deduce that every normal channel on $\mathcal{B}(H_0)$ has the form

$$\Phi(x) = V^*(x \otimes 1_E)V$$

and that its preadjoint is a Schrödinger-picture channel.

A positive functional ψ is **normal** if it preserves suprema of bounded increasing nets of positive elements and **faithful** if $\psi(a^*a) = 0$ implies $a = 0$. A **type II₁ factor** is an infinite-dimensional factor with a faithful normal tracial state

$$\tau : M \rightarrow \mathbb{C}, \quad \tau(ab) = \tau(ba), \quad \tau(1) = 1.$$

For projections $p, q \in M$, write $p \sim q$ if there is $v \in M$ such that

$$v^*v = p, \quad vv^* = q.$$

Write $p \precsim q$ if p is equivalent to a projection r satisfying $rq = r$.

Exercise 97 (Projection dimension)

Let M be a type II_1 factor with normalized trace τ . Prove that

$$p \sim q \implies \tau(p) = \tau(q),$$

and prove the comparison theorem

$$p \precsim q \iff \tau(p) \leq \tau(q).$$

Deduce that $\tau(p) = \tau(q)$ if and only if $p \sim q$.

3.10 Conditional expectations and subfactors

A von Neumann algebra is **finite** if $v^*v = 1$ implies $vv^* = 1$. Let $N \subseteq M$ be finite von Neumann algebras with a common faithful normal tracial state τ . A **trace-preserving conditional expectation** is a normal unital completely positive map $E_N : M \rightarrow N$ such that

$$\tau \circ E_N = \tau, \quad E_N(n_1 x n_2) = n_1 E_N(x) n_2$$

for $x \in M$ and $n_1, n_2 \in N$. Let $L^2(M, \tau)$ be the completion of M for

$$\langle x, y \rangle_\tau = \tau(y^* x).$$

Let M act on $L^2(M, \tau)$ by left multiplication.

Exercise 98 (Norm-one projections are conditional expectations)

Let A be a unital C^* -algebra, $B \subseteq A$ a unital C^* -subalgebra, and $P : A \rightarrow B$ a linear projection with $\|P\| = 1$. Prove Tomiyama's theorem [21]: P is positive, completely positive, and

$$P(b_1 a b_2) = b_1 P(a) b_2.$$

Exercise 99 (Conditional expectations give Jones projections)

Prove that there is a unique trace-preserving conditional expectation $E_N : M \rightarrow N$ and that it extends to the orthogonal projection

$$e_N : L^2(M, \tau) \longrightarrow L^2(N, \tau).$$

Prove that

$$e_N = e_N^* = e_N^2, \quad e_N x e_N = E_N(x) e_N$$

for every $x \in M$.

The **basic construction** is

$$M_1 = \langle M, e_N \rangle'' \subseteq \mathcal{B}(L^2(M, \tau)),$$

the double commutant of the algebra generated by M and e_N . For a projection $p = [p_{ij}] \in M_k(N)$ acting by left multiplication, define

$$\dim_N(p L^2(N, \tau)^k) = \sum_{i=1}^k \tau(p_{ii}).$$

An inclusion $N \subseteq M$ has **finite Jones index** if $L^2(M, \tau)$ is isomorphic as a right N -module to such a module; then

$$[M : N] = \dim_N L^2(M, \tau).$$

Exercise 100 (The Pimsner–Popa bound detects finite index)

For an inclusion $N \subseteq M$ of type II_1 factors with trace-preserving expectation E_N , prove that finite index is equivalent to the existence of $\lambda > 0$ satisfying [22]

$$E_N(x) \geq \lambda x \quad (x \in M_+).$$

Prove that the largest such λ is $[M : N]^{-1}$ and interpret this order inequality as a quantitative strengthening of complete positivity.

Exercise 101 (Jones basic construction)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that M_1 is a finite factor and that its normalized trace satisfies [23]

$$\tau_1(xe_N y) = \frac{1}{[M : N]} \tau(xy)$$

for $x, y \in M$.

The **Jones tower** is the sequence obtained by iterating the basic construction:

$$N = M_{-1} \subset M = M_0 \subset M_1 \subset M_2 \subset \cdots.$$

Let e_i be the Jones projection used to construct M_i from M_{i-1} , and set

$$\delta = \sqrt{[M : N]}.$$

The **Temperley–Lieb algebra** with loop parameter δ is generated by U_i subject to

$$U_i^2 = \delta U_i, \quad U_i U_j = U_j U_i \quad (|i - j| \geq 2), \quad U_i U_{i \pm 1} U_i = U_i.$$

Exercise 102 (Temperley–Lieb relations)

Prove that the Jones projections satisfy [26, 24]

$$e_i^2 = e_i = e_i^*, \quad e_i e_j = e_j e_i \quad (|i - j| \geq 2),$$

$$e_i e_{i \pm 1} e_i = \delta^{-2} e_i.$$

Deduce that $U_i \mapsto \delta e_i$ defines a representation of the Temperley–Lieb algebra.

Exercise 103 (Braid representation)

Let $q \in \mathbb{C}^\times$ satisfy $\delta = q + q^{-1}$ and define $\sigma_i = q 1 - U_i$. Prove that σ_i is invertible with inverse $q^{-1} 1 - U_i$, and prove that

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1},$$

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2).$$

Deduce that the σ_i define a representation of the braid group [25].

A **Markov trace** on the Temperley–Lieb tower is a compatible family of positive normalized traces satisfying

$$\mathrm{tr}(xU_i) = \delta^{-1} \mathrm{tr}(x)$$

whenever x belongs to the algebra generated by the preceding generators.

Exercise 104 (Positivity quantizes subfactor indices below four)

Define

$$P_0(\delta) = 1, \quad P_1(\delta) = \delta, \quad P_{n+1}(\delta) = \delta P_n(\delta) - P_{n-1}(\delta).$$

Prove that the normalized traces on the Jones tower restrict to a Markov trace.

Prove that, when $\sin \theta \neq 0$,

$$P_n(2 \cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}.$$

Use positivity of the Markov trace to prove that if $0 < \delta < 2$, then

$$\delta = 2 \cos \left(\frac{\pi}{m} \right)$$

for some integer $m \geq 3$.

Exercise 105 (Jones index theorem)

Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that [24]

$$[M : N] \in \left\{ 4 \cos^2 \left(\frac{\pi}{n} \right) : n \geq 3 \right\} \cup [4, \infty).$$

Relate the discrete values to the $SU(2)_k$ braid representations and Chern–Simons Wilson-loop invariants constructed above.

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